

HDOC Day-42

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Batch -15

1. Algorithm

Main Algorithm

1. Start.
2. Display the menu:
 - Remove all vowels from a string.
 - Find the first repeating lowercase alphabet.
 - Check whether two strings are anagrams.
 - Exit.
3. Read the user's choice.
4. Use **switch** to call the corresponding function.
5. Repeat the menu until the user chooses Exit.
6. Stop.

Function 1: Remove All Vowels

1. Read a string.
2. Traverse the string character by character.
3. Check whether each character is **a, e, i, o, u** or its uppercase equivalent.
4. If it is not a vowel, copy it to the resulting string.
5. Add the null character '**\0**'.
6. Display the string after removing vowels.

Function 2: First Repeating Lowercase Alphabet

1. Read a string.
2. Create a frequency/visited array of size 26 initialized to zero.
3. Traverse the string from left to right.
4. For every lowercase alphabet (**a** to **z**):
 - If it has already appeared, it is the first repeating lowercase alphabet.
 - Otherwise, mark it as seen.
5. If no lowercase alphabet repeats, display an appropriate message.

Function 3: Check Anagrams

1. Read two strings.
2. If their lengths are different, they cannot be anagrams.
3. Create two frequency arrays of size 256 initialized to zero.
4. Count the frequency of every character in both strings.
5. Compare the frequency arrays.

6. If all frequencies are equal, the strings are anagrams.
7. Otherwise, they are not anagrams.

2. C Program

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
#define SIZE 200
```

```
void removeVowels(void);
```

```
void firstRepeatingLowercase(void);
```

```
void checkAnagram(void);
```

```
int main(void)
```

```
{
```

```
    int choice;
```

```
    do
```

```
    {
```

```
        printf("\n===== MENU =====\n");
```

```
        printf("1. Remove all vowels from a string\n");
```

```
        printf("2. Find first repeating lowercase alphabet\n");
```

```
        printf("3. Check if two strings are anagrams\n");
```

```
printf("4. Exit\n");

printf("=====\n");

printf("Enter your choice: ");

scanf("%d", &choice);

getchar();


switch (choice)

{

    case 1:

        removeVowels();

        break;


    case 2:

        firstRepeatingLowercase();

        break;


    case 3:

        checkAnagram();

        break;


    case 4:

        printf("Exiting the program...\n");
```

```
break;
```

```
default:
```

```
printf("Invalid choice! Please enter 1 to 4.\n");
```

```
}
```

```
} while (choice != 4);
```

```
return 0;
```

```
}
```

```
/* Function to remove all vowels */
```

```
void removeVowels(void)
```

```
{
```

```
char str[SIZE];
```

```
char result[SIZE];
```

```
int i, j = 0;
```

```
printf("Enter a string: ");
```

```
fgets(str, SIZE, stdin);
```

```
str[strcspn(str, "\n")] = '\0';
```

```

for (i = 0; str[i] != '\0'; i++)
{
    char ch = tolower((unsigned char)str[i]);

    if (ch != 'a' && ch != 'e' && ch != 'i' &&
        ch != 'o' && ch != 'u')
    {
        result[j] = str[i];

        j++;
    }
}

result[j] = '\0';

printf("String after removing vowels: %s\n", result);
}

```

/* Function to find first repeating lowercase alphabet */

```
void firstRepeatingLowercase(void)
```

```

{
    char str[SIZE];

```

```
int seen[26] = {0};

int i;

char repeating = '\0';


printf("Enter a string: ");

fgets(str, SIZE, stdin);


str[strcspn(str, "\n")] = '\0';


for (i = 0; str[i] != '\0'; i++)
{
    if (str[i] >= 'a' && str[i] <= 'z')
    {
        int index = str[i] - 'a';

        if (seen[index] == 1)
        {
            repeating = str[i];

            break;
        }

        seen[index] = 1;
```

```
}  
  
}
```

```
if (repeating != '\0')  
  
    printf("First repeating lowercase alphabet: %c\n",  
  
        repeating);  
  
else  
  
    printf("No repeating lowercase alphabet found.\n");  
  
}
```

```
/* Function to check whether two strings are anagrams */
```

```
void checkAnagram(void)
```

```
{
```

```
    char str1[SIZE], str2[SIZE];
```

```
    int freq1[256] = {0};
```

```
    int freq2[256] = {0};
```

```
    int i, flag = 1;
```

```
    printf("Enter first string: ");
```

```
    fgets(str1, SIZE, stdin);
```

```
    str1[strcspn(str1, "\n")] = '\0';
```

```

printf("Enter second string: ");

fgets(str2, SIZE, stdin);

str2[strcspn(str2, "\n")] = '\0';

if (strlen(str1) != strlen(str2))

{

    printf("The strings are not anagrams.\n");

    return;

}

for (i = 0; str1[i] != '\0'; i++)

{

    freq1[(unsigned char)str1[i]]++;

    freq2[(unsigned char)str2[i]]++;

}

for (i = 0; i < 256; i++)

{

    if (freq1[i] != freq2[i])

    {

        flag = 0;

        break;

```

```

    }

}

if (flag)

    printf("The strings are anagrams.\n");

else

    printf("The strings are not anagrams.\n");

}

```

3. Test-Case Table

Test	Choice	Input	Expected Output
1	1	Hello World	Hll Wrld
2	1	AEIOUaeiou	Empty string
3	1	Computer Science	Cmptr Scnc
4	2	programming	r
5	2	abcdef	No repeating lowercase alphabet
6	2	hello	l
7	3	listen,silent	Anagrams
8	3	earth,heart	Anagrams
9	3	hello,world	Not anagrams
10	5	—	Invalid choice
11	4	—	Exit

Note: In this implementation, anagram checking is **case-sensitive and space-sensitive**. For example, **Listen** and **silent** are not treated as anagrams.

4. Sample Execution

```
[aaryanrajput@aaryans-macbook ~ % nano day42.c
[aaryanrajput@aaryans-macbook ~ % clang day42.c -o day42.c
[aaryanrajput@aaryans-macbook ~ % ./day42.c

===== MENU =====
1. Remove all vowels from a string
2. Find first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit
=====
Enter your choice: 1
Enter a string: hello world
String after removing vowels: hll wrld

===== MENU =====
1. Remove all vowels from a string
2. Find first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit
=====
Enter your choice: 2
Enter a string: hello world
First repeating lowercase alphabet: l

===== MENU =====
1. Remove all vowels from a string
2. Find first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit
=====
Enter your choice: 3
Enter first string: hello world
Enter second string: hello computer
The strings are not anagrams.

===== MENU =====
1. Remove all vowels from a string
2. Find first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit
=====
Enter your choice: 4
Exiting the program...
aaryanrajput@aaryans-macbook ~ %
```